Fine-tuning Seq2Seq Language Models (T5) for Topic Labeling

A Transfer Learning Approach

Problem Statement

Challenges in Topic Labeling

- **Output Gap**: Topic modeling (e.g., LDA) produces lists of keywords that require human interpretation.
- **Manual Annotation**: Time-consuming, subjective, and labor-intensive.
- **LLM API Limitations**:
  - High inference costs and latency.
  - Unpredictable stochasticity and potential hallucination.
  - Lack of fine-grained control and privacy concerns.

Proposed Solution

Specialized Seq2Seq Fine-tuning

- **Approach**: Fine-tune T5 models specifically for topic labeling.
- **Goal**: Combine the semantic linguistic understanding of pretrained transformers with task-specific adaptation.
- **Benefits**:
    - **Controllable**: Deterministic and reproducible outputs.
    - **Efficient**: Offline, cost-free inference after training.

LLM Comparison

Establishing a Reliable Data Source

- **LLM Benchmarking**: Evaluated six models (GPT-4, Claude, Perplexity, Gemini Pro, Grok, DeepSeek) using SPECTER semantic similarity on 67 pairs from our previous work.
- **Results**: Grok (0.887) and Gemini Pro (0.885) achieved the highest mean similarity between confirmed label and generated label.
- **Selection**: Gemini Pro was chosen for large-scale data generation due to its integration with Google Sheets.

Expert Validation Study I

Validating LLM-Generated Training Data

- **Hypothesis**: LLMs can generate high-quality keyword-label pairs with expert-level accuracy.
- **Participants**: 35 PhD holders across 8 different faculties.
- **Traps** were embedded to avoid “Agree to all” responses.
  - 2 participants were trapped.
  - **All** their votes will not be considered in the collected dataset

Expert Validation Study II

Pairs Generation (topic_word-topic-label)

- For every participant, his/her specialized research fields were determined through Google Scholar profiles.
  - Example: Dr. Maggie = {AI, Cybersecurity, Cloud Computing, Data Science, Computer Engineering}
- Each research field was broken down into **7** topic pairs
  - Example: Data Science = {Core Machine Learning, Big Data Technologies, Data Preprocessing Techniques + 4 more}
- **35** Professors → **161** research fields → **1127** pairs.

Prompt

You are an expert research scientist, with a deep focus on [Your Research Field]. Your task is to generate 7 unique (topic_words, topic_label) pairs relevant to this field. Follow these strict guidelines: 1. `topic_words`: This must be a JSON list of 5 lowercase, single-word strings. These words should be highly related keywords, as if they were the top 5 tokens from an LDA topic model. 2. `topic_label`: This must be a concise, human-readable string (2-5 words) that accurately summarizes the `topic_words`. Here are examples from a *different* field (Computational Biology) to show the exact format: [{ "topic_words": ["gene", "expression", "rna-seq", "transcriptome", "differential"], "topic_label": "Gene Expression Analysis" }, { "topic_words": ["protein", "structure", "folding", "docking", "molecular"], "topic_label": "Protein Structure & Docking" }, { "topic_words": ["crispr", "cas9", "editing", "genome", "mutation"], "topic_label": "CRISPR Genome Editing" }] Now, generate 7 new pairs specifically for my field: * **Sub-discipline:** [Your Sub-discipline] * **Research Field:** [Your Research Field] Produce the output as a single, valid JSON list of objects.

Expert Validation Study III

Results:

- **30** participants actually responded in time (28 excluding trapped).
- **Agreement Rate**: 87.22% among experts.
- **Statistical Significance**: One-sample z-test yielded $p < 0.00001$, confirming the methodology is significantly better than random chance.
- **812** verified pairs out of 931 valid votes.

Datasets Construction I

Two Complementary Training Datasets

- **Verified Dataset**:
  - 812 high-quality pairs.
  - Strictly expert-validated for accuracy.
- **Taxonomy Dataset**:
  - 6,377 pairs spanning 911 research fields.
  - Based on Elsevier's Digital Commons Three-Tiered Taxonomy.
  - Provides broad coverage and diversity across scientific disciplines.

Datasets Construction II

Preparing the Test Set

- Already prepared pairs for professors who haven’t submitted the form in time.
- 5 professors: 2 from MET, 2 from IET, and 1 from Physics.
- Results were printed and handed to Dr. Maggie & Marwa (on behalf of Dr. Mervat) in our last meeting, for human evaluation.

Model Architecture & Strategy

Fine-tuning T5 Models

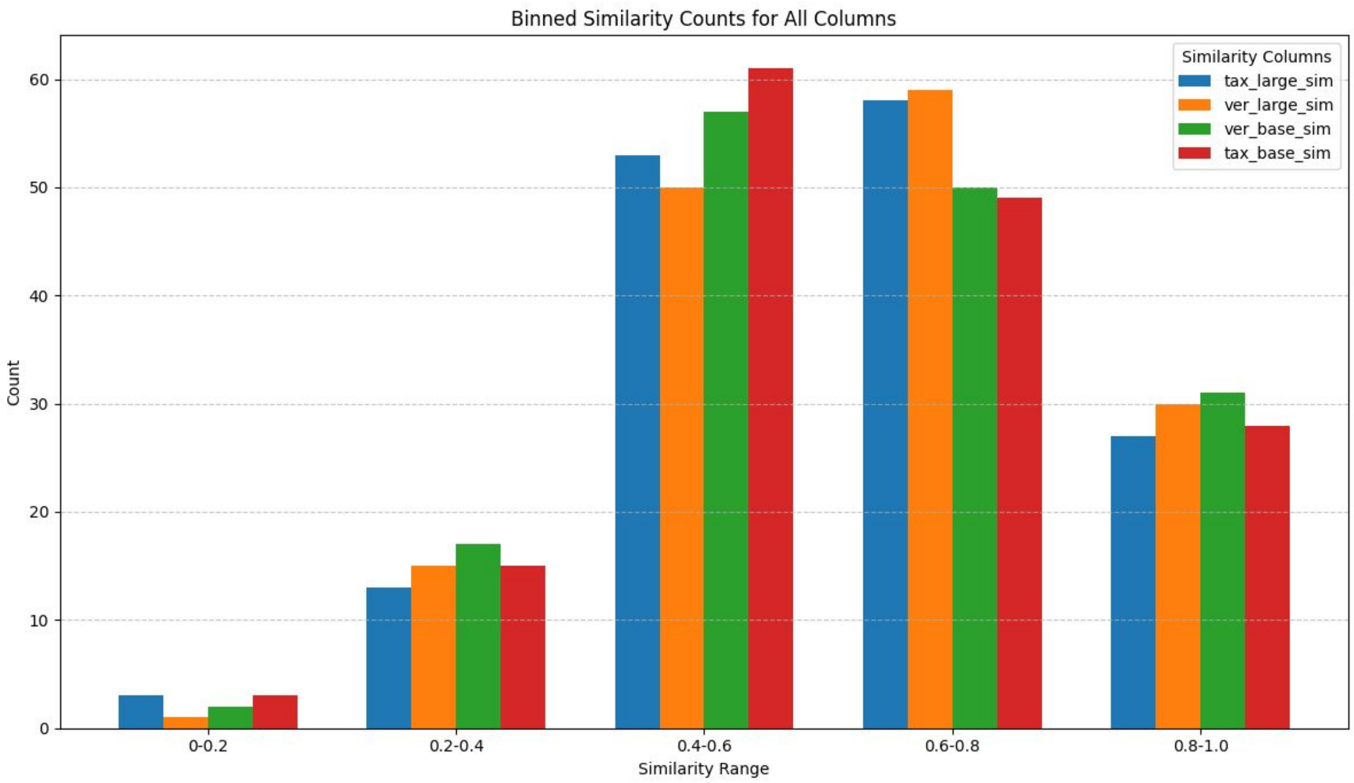
- **Architectures**: T5-base (220M) and T5-large (770M).
- **Encoder-Freezing Strategy**:
  - First four encoder layers remain frozen during training.
  - **Purpose**: Preserves general linguistic features while allowing higher layers to adapt to the specific task.
- *Beam search, Random Dropout, Order Shuffling* for better **Generalization**.

Experimental Evaluation

Model	Mean Cosine Similarity
Verified Base	0.628
Verified Large	0.644
Taxonomy Base	0.627
Taxonomy Large	**0.645**

- **Metric**: Cosine similarity using SPECTER embeddings between topic_words and generated label.
- **Key Finding**: Larger model capacity (T5-large) consistently shifts predictions from acceptable to higher quality.
- Human Evaluation needed.

Labeler Models Comparison



Error Analysis & Limitations

Current Challenges

- **Acronym Handling**: Systematic failure in interpreting technical abbreviations (e.g., "SHAP" and "LIME" misinterpreted as “shape” and lime 🍋).
- **Causes**:
    1. Subword tokenization splits acronyms into meaningless characters.
    2. Pretraining bias toward common vocabulary over technical terms.
- **Suggested Fixes**: Acronym-aware tokenization and glossaries.